

# Clockwork — Plain-English Product Specification

## MASTER PRODUCT SPECIFICATION:

This document serves as the plain-English product specification for Clockwork. It defines the app's core concepts, behavioral scoring rules, visual aesthetics, user workflows, and features without technical jargon.

## 1. App Philosophy & Visual Style

**Clockwork** is a minimalist, privacy-focused iOS app designed to help users stay aligned with a structured daily routine. Unlike typical habit trackers that rely on binary pass/fail streaks, Clockwork measures **precision, context, and consistency**.

### Visual Aesthetic (Pure Minimalist Terminal)

- **Background:** True OLED Black (#000000).
- **Typography:** Pure monospaced lettering across the entire interface (no standard rounded UI fonts).
- **Styling & Layout:** Uses ASCII-style borders, text dividers (+---, |, .---), and bracketed action buttons (e.g., [ MARK DONE ]).
- **Color Accenting:** High-contrast monochromatic text (white, gray, dark gray) paired with subtle, single-color accents assigned to specific habit categories or scoring indicators.

## 2. The Core Telemetry Engine (3 Behavioral Rings)

Instead of scoring habits on a simple "did it or didn't do it" basis, the app evaluates routines through three distinct scoring mechanisms:

### 1. The Clock Ring (CLK - Temporal Precision):

- Measures how close to the scheduled start time a habit was performed.
- **Strict Timing Habits:** Must be started within a tight ~30-minute window of the target time to get full credit.
- **Flexible Timing Habits:** Have a broad ~180-minute window around the target time.
- **Early vs. Late Asymmetry:** Logging a habit early is treated leniently (e.g., up to 40 minutes early gets full credit for strict habits; any time early gets full credit for flexible habits). Logging a habit late immediately begins decaying the score.

### 2. The Chain Ring (CHN - Contextual Habit Stacking):

- Tracks habits that are tied to another anchor habit (e.g., starting a workout right after waking up).
  - Measures the time gap between when the *first* habit finished and when the *second* habit started, completely ignoring what time of day it is on the clock.
  - **Parent Grace Rule:** If an anchor/parent habit is skipped, the dependent child habit automatically detaches or defaults to baseline credit so the user's chain score isn't unfairly penalized.
3. **The Show-Up Ring (SHW - Pure Execution):**
- Measures raw completion consistency regardless of timing or sequence.
  - For habits designated with a "Show-Up Bonus", completing the habit guarantees a baseline 50% credit instantly, with the remaining 50% scaling based on how close to the target time it was completed.

### 3. User Account Isolation & Multi-User Support

The app supports multi-device sync and account isolation via Sign in with Apple. Every habit, log entry, and setting is explicitly bound to the authenticated user account to ensure complete data privacy and isolation across devices or cloud backups.

### 4. Habit Entry & Execution Options

Users are never forced to hit "Start" and "Stop" live for every habit. The app supports four flexible logging workflows:

1. **One-Tap Instant Log (Default):** Tapping "Mark Done" (on the main screen or widget) logs the completion at the current exact time and automatically calculates the duration using the habit's pre-set target length.
2. **Retrospective Editing:** If logging hours later, users can adjust their start time or duration using quick monospaced step buttons ([**-**] and [**+**]) in 5-minute increments.
3. **Reset to Target Macro:** A single-tap [ **RESET TO TARGET** ] button instantly populates actual times to match the exact scheduled target for fast cleanup.
4. **Siri Voice Commands:** Hands-free background logging using voice commands such as *"Hey Siri, mark morning commute done in Clockwork"* or *"Log water in Clockwork"*.

### 5. Pause Engine (Hikes, Treks & Holidays)

Users can pause tracking for any date by selecting a reason (Hike, Trek, Holiday, or Other).

- When a date is paused, all habits for that day are excluded from calculations.

- Paused days do not lower historical trend lines or break long-term execution averages.

## 6. Main Dashboard & Home Screen Widget

### Timeline Dashboard

- Displays the current date's habits in a flat, chronological list with ASCII timeline headers.
- Shows target time vs. actual time, time variance in minutes, current execution status, and inline action controls ([ MARK DONE ] or step adjusters).
- Includes a top status bar displaying overall summary metrics for the three rings (CLK, CHN, SHW).

### Interactive Home Screen Widget (Single-Focus Context Window)

- A high-density widget that highlights the **single active or upcoming habit**.
- Displays the task name (supporting up to 4 lines of clean text wrapping), target time, and a direct [ MARK DONE ] interactive button.
- Displays current CLK, CHN, and Plugin/Show-Up summary scores embedded directly in the lower status bar.